Higher Diploma in Computer Science

Computer Systems & Networks

SPRINGBOARD+ www.springboardcourses.ie
Ireland's European Structural and Investment Funds Programmes 2014-2020
Co-funded by the Irish Government and the European Union
An Roinn Oideachais agus Eolaíochta | Department of Education and Science
HEA HIGHER EDUCATION AUTHORITY AN tÚDARÁS um ARD-OIDEACHAS
European Union Investing in your future European Social Fund
SETU Ollscoil Teicneolaíochta an Oirdheiscirt South East Technological University

Contact Me

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slack

Weekly Webinar Schedule

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	
10:45						10:45
12:15	Computer Systems *Webinar* 12:15-2:00		Computer Systems *Webinar* 12:15-2:00	Database *Webinar* 2:15-2:00		12:15
2:00						13:45

Expectations

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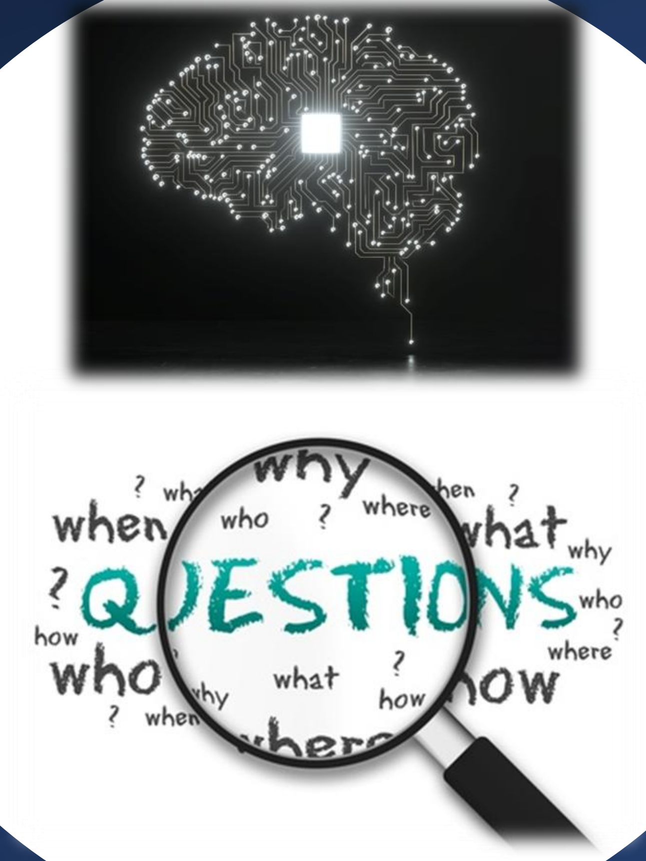
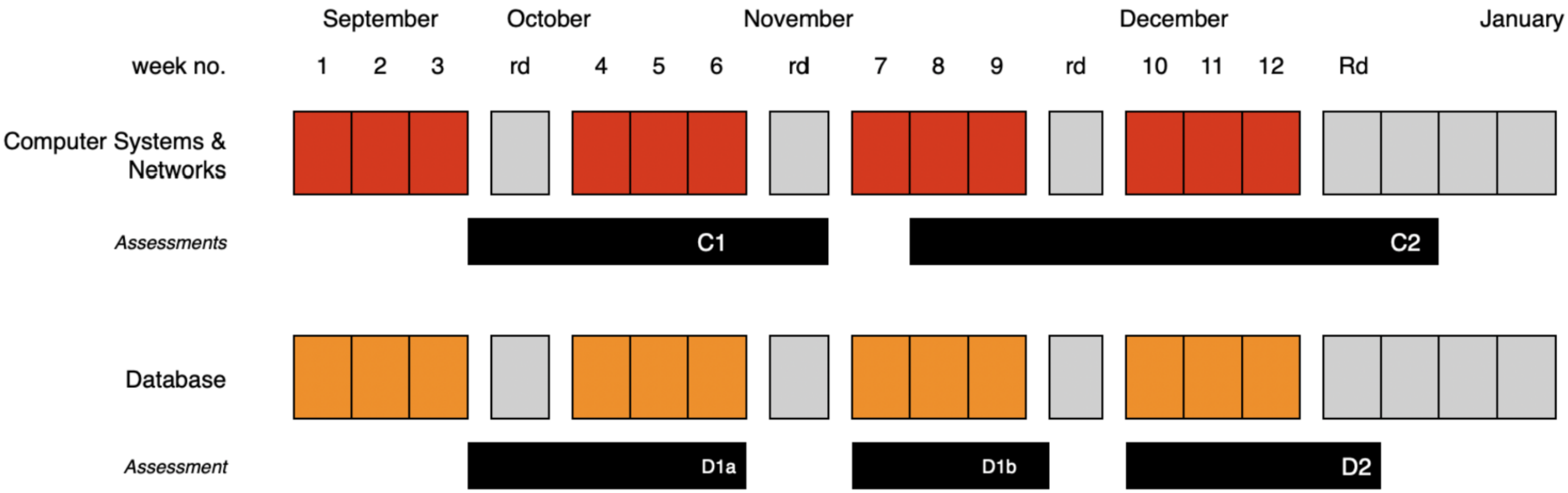
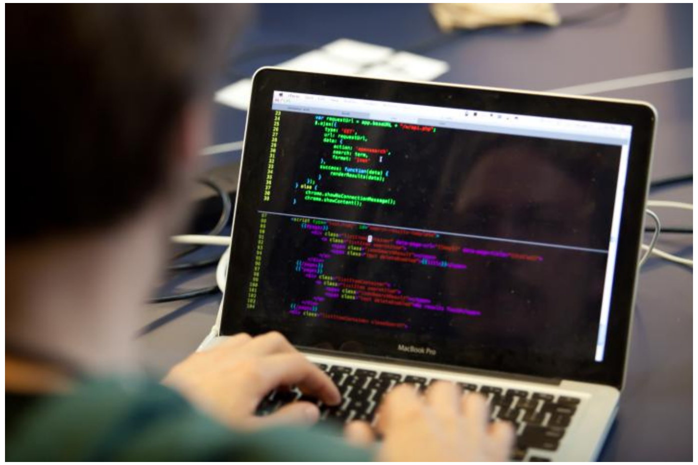


Semester 2 Assessment Schedule



TBC CA for Caroline’s Computer Systems

5% Cisco NetAcad chapter exams (*due end Oct*)



25% Linux Shell scripting

5% InClass Exercises

CA for Caroline’s Computer Systems

5% Cisco NetAcad chapter exams (*ongoing*) cisco Networking Academy

YOU SHOULD HAVE RECEIVED AN EMAIL INVITE FROM ME, SIMILAR TO:

Cisco Networking Academy ☆ **Join Linux Essentials at Waterford Institute of Technolo...** Hello

Module 1 – Introduction to Linux

Chapter 1

Module 2 – Operating Systems

Chapter 2

Chapter 02 Exam

Module 3 – Working in Linux

Chapter 3

Chapter 03 Exam

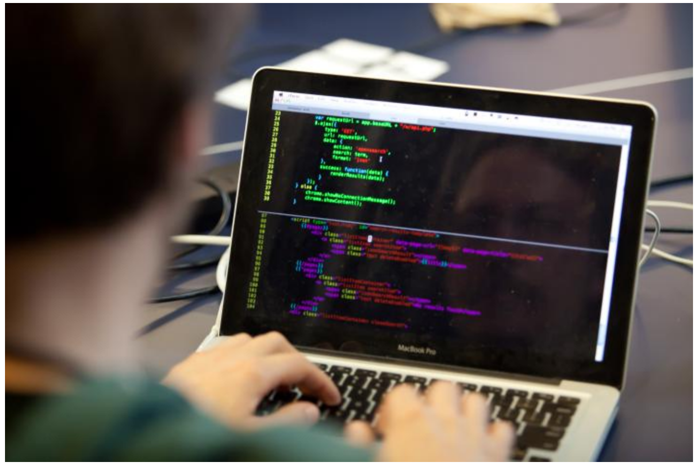
Module 4 – Open Source Software and Licensing

Chapter 4

Chapter 04 Exam

CA for Caroline’s Computer Systems

5% Cisco NetAcad chapter exams (*due end Oct*)

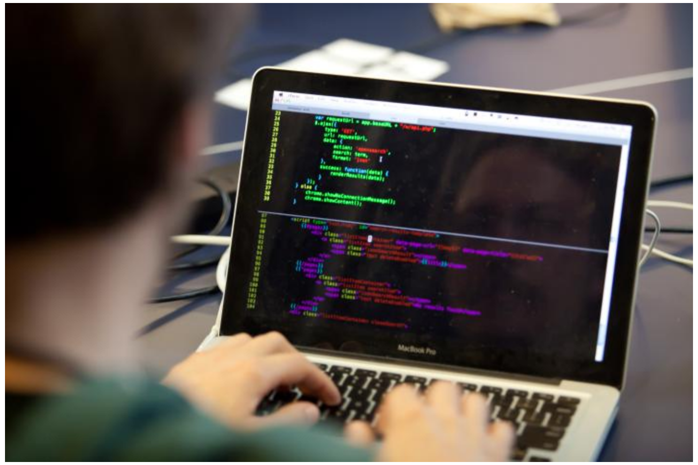


25% Linux Shell scripting assessment (Given *end Sept, Due END Oct ‘25*)

5% InClass Exercises

CA for Caroline's Computer Systems

5% Cisco NetAcad chapter exams (*due end Oct*)

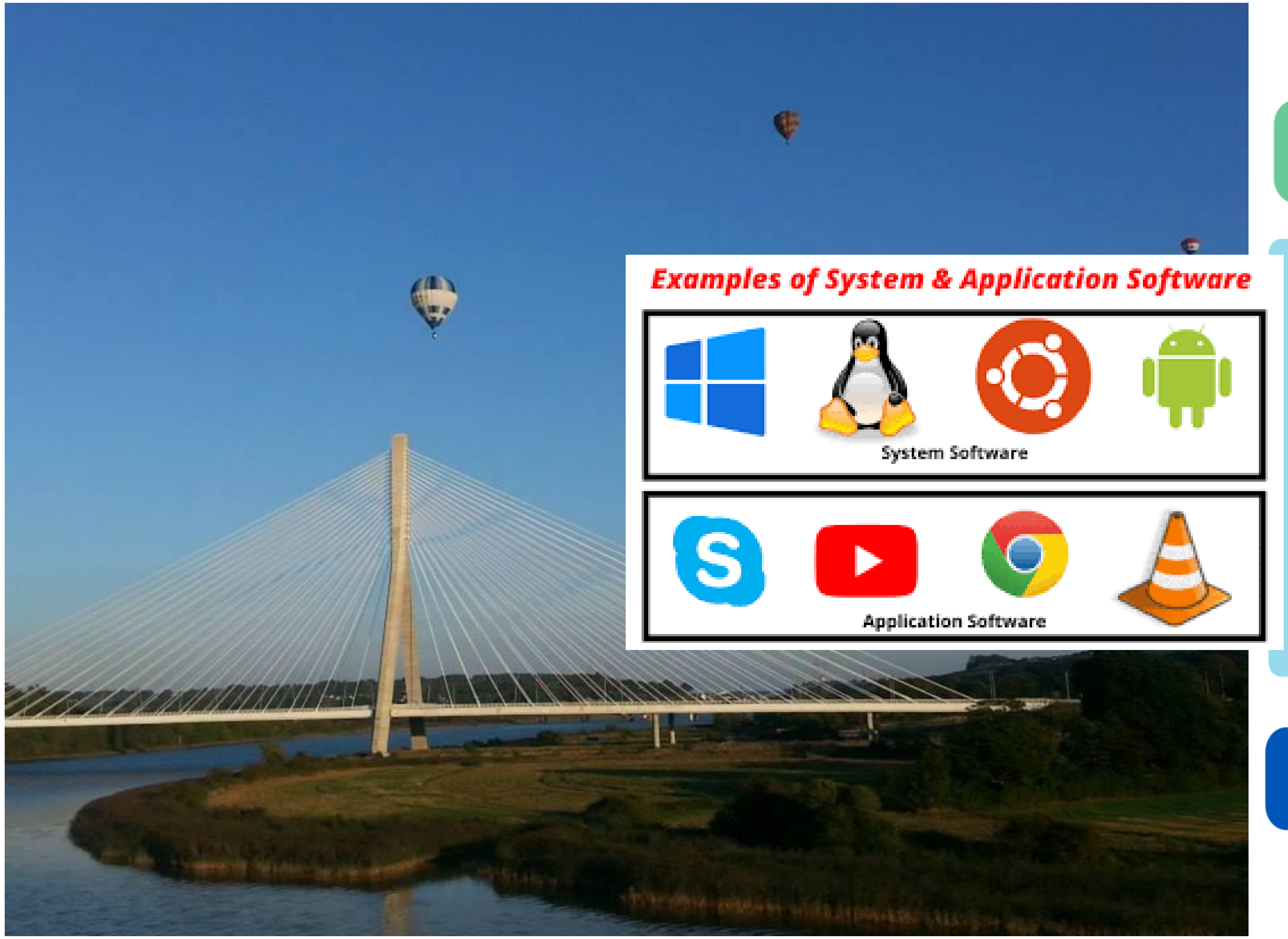


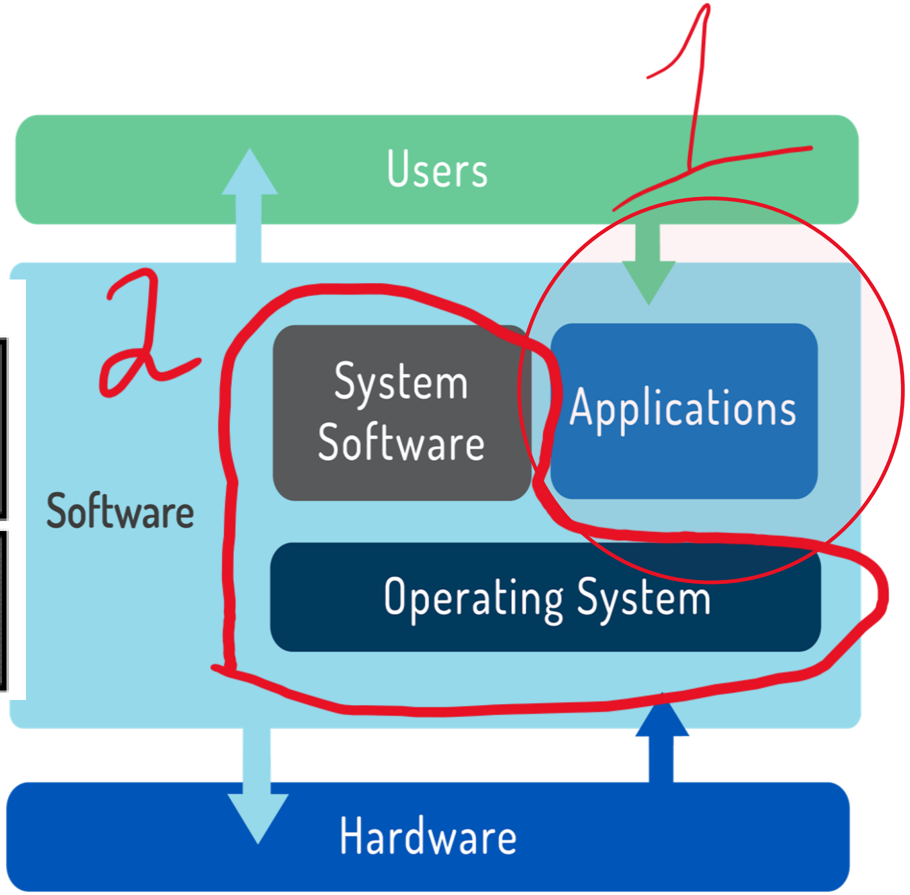
25% Linux Shell scripting

5% Classwork: completed scripts (during Sept)

Introduction to Operating Systems

Operating System (abbr. OS)





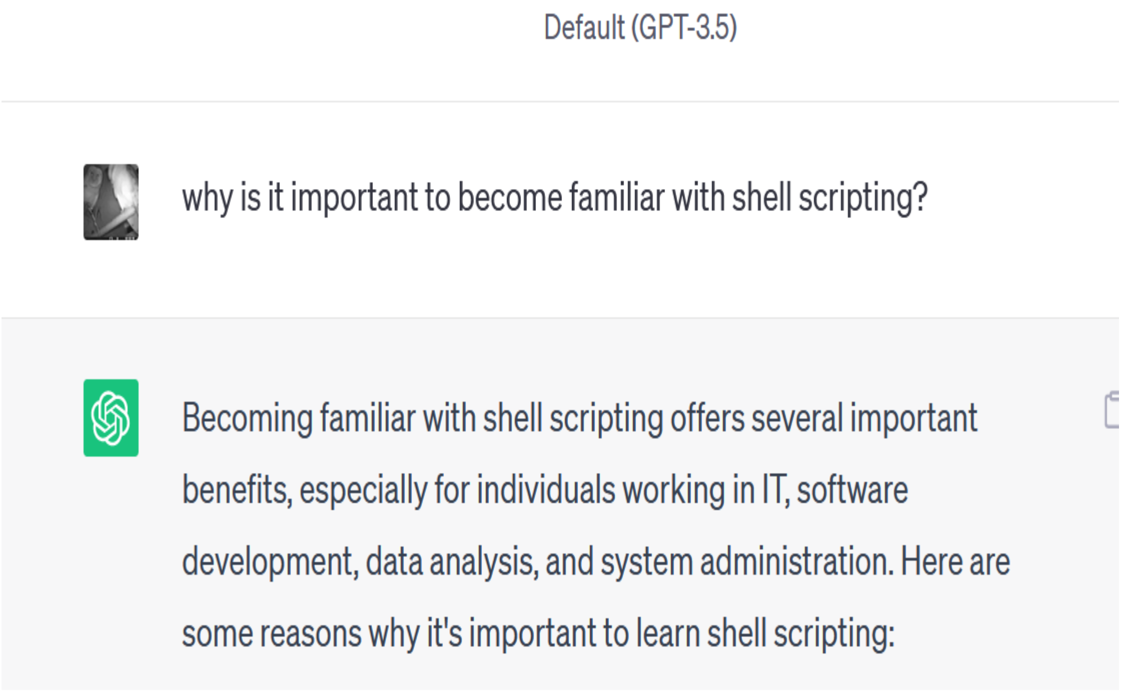
Decision Points when Considering which OS is best

- **Role**: Accessed by 1 user directly (desktop) or many users remotely (server)?
- **Function**: Does it need to run specific software?
- **Life Cycle**: What is the service lifetime?
- **Stability**: Are OS releases *beta* (not tested “in the wild”) or *stable* (tested)?
- **Compatibility**: Is it backwards compatible
- **Cost**: important factor for new systems

Introduction to Linux (as an OS)

Linux is Everywhere!

- Linux is everywhere, Linux jobs are everywhere, there is a shortage of Linux talent, and, Linux skills are in demand in just about every industry and job category.



Linux is Open Source

- The development of Linux closely parallels the rise of *open source software*.
- The open source philosophy is that

You have a right to obtain the software **source code** and

You have a right to **modify** it for your own use.

Linux is around a long time

- Linux started in 1991 as a hobby project by Linus Torvalds, a Finnish born computer scientist studying at the University of Helsinki.


ZDNet

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MUST READ: How COVID-19 pandemic, Amazon made retailers fundamentally stronger

Linux turns 30: Linus Torvalds on his "just a hobby" operating system

It's been 30 years since Finnish graduate student Linus Torvalds drafted a brief note saying he was starting a hobb
operating system. The world would never be the same.

By Steven J. Vaughan-Nichols for Linux and Open Source | August 25, 2021 -- 16:53 GMT (17:53 BST) | Topic: Enterprise Software

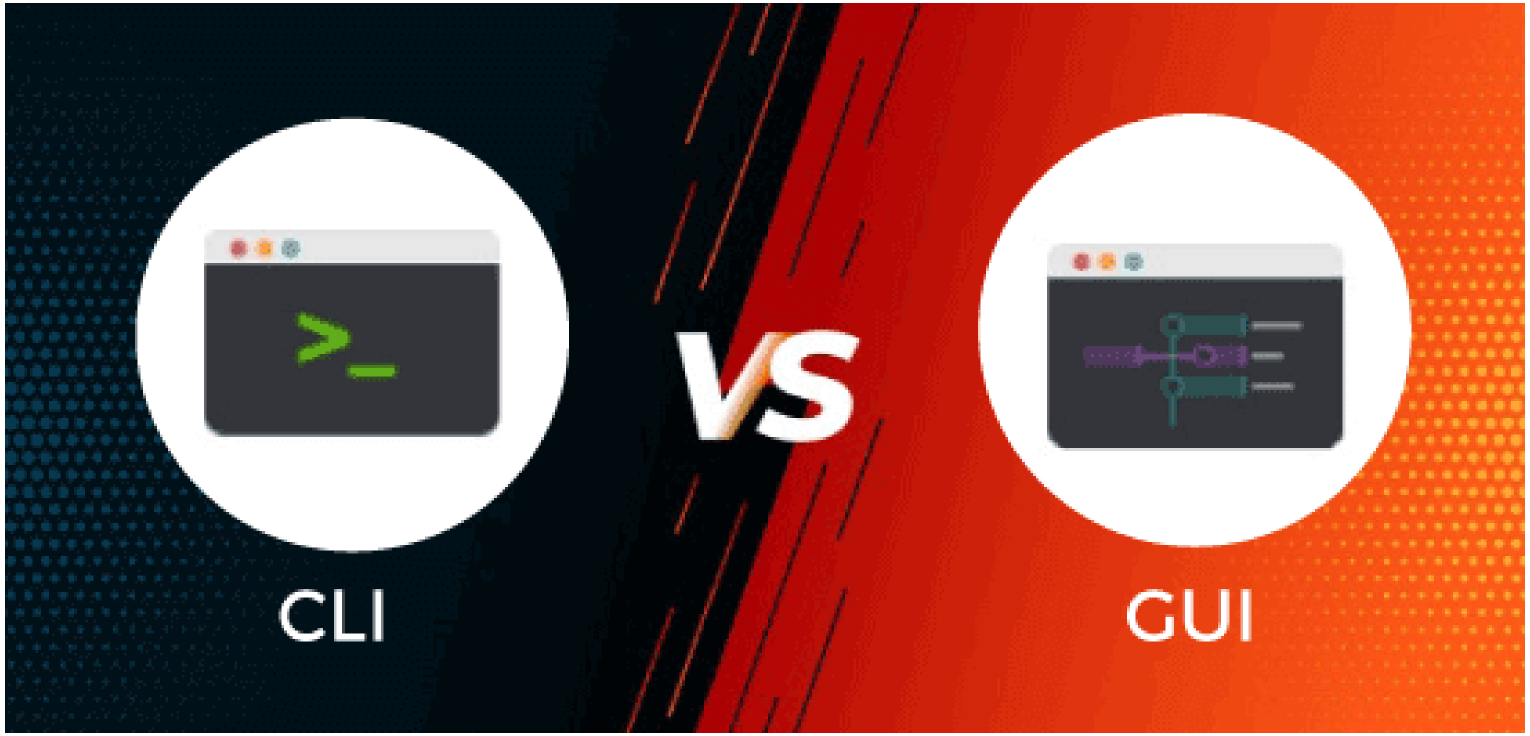
Linux is a Kernel

- Linux means the *kernel* of the system, which is the central controller of everything that happens on the computer.

- Linux is a combination of software called **GNU/Linux**, which defines the *operating system*.
  - **GNU** is the *free software* that provides open source equivalents of many common UNIX commands.
  - The **Linux** part of this combination is the *Linux kernel*, which is the core of the operating system.

Linux Embraces the CLI

Two basic types of interfaces available that allow you to interact with the operating system



Linux Distros

A distribution refers to the Linux kernel, tools, and suite of applications that come bundled together.

Linux distros refer to an OS composed of open-source programs designed by numerous developers and programmers.

Lubuntu = Lightweight Ubuntu

- Ubuntu is one of the popular Linux distributions available for desktops and laptops
- A new version of Ubuntu is released every April and October.
- There's a thriving Ubuntu community online

  - Refer back to our [May ‘24 onsite Prep Lab]

~~Linux~~ YOU Embrace the CLI

- Enter the Command

$cal

- to view the built in calendar.

```
compsys@compsys-virtualbox:~$ cal
    August 2023
Su Mo Tu We Th Fr Sa
       1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
```

```
compsys@compsys-virtualbox:~$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
```

$sudo apt install <<program>>

```
compsys@compsys-virtualbox:~$ cal sept 2024
   September 2024
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30
```

$sudo apt update

Use this command to update the package database any time

~~Linux~~ YOU Embrace the CLI

- Try look up the current weather in Kilkenny using

$ansiweather –l *your_town_name*

```
compsys@compsys-virtualbox:~$ ansiweather -l kilkenny
 Weather in Kilkenny: 18 °C - UVI: 5.32 - Wind: 4.02 m/s WNW - Humidity: 75% - Pressure: 1016 hPa
```

Be familiar with the help **man**ual available

What does the **–l** stand for in the command line instruction

$ansiweather **–l** kilkenny

~~Linux~~ YOU Embrace the CLI

- Uninstall ansiweather

$sudo apt-get remove *program-name*

~~Linux~~ YOU Embrace the CLI

- Try out: **$cowsay Well hello there September!**

Do you need to install it?

- Once finished, **uninstall**

Don't like cows?? $cowsay –l

$cowsay –f turkey Almost 100 days to Christmas, gobble!

These are examples of optional arguments that many CLI commands accept

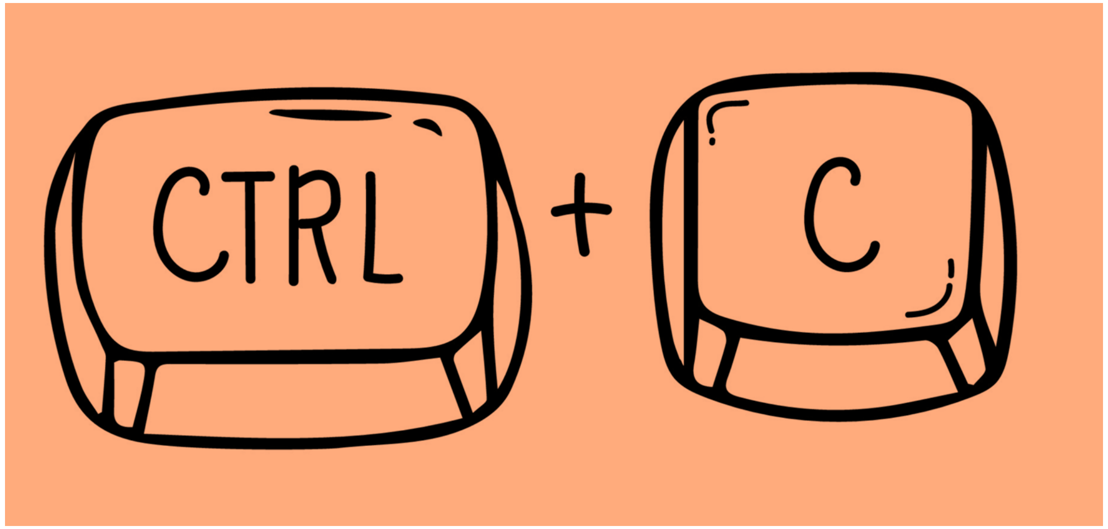
~~Linux~~ YOU Embrace the CLI

- Try out: **$cmatrix**

Now **$figlet Lets go**

And finally, what happens here **$figlet Let’s go**

Why?



Try out this weeks Lab